HYDROTECH - ERD RELATIONSHIPS

Diagram reference - all foreign keys and cardinality rules

R1 profile.updated_by → users.user_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R2 settings.updated_by → users.user_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R3 dithane_schedules.updated_by → users.user_id

At most one schedule in the entire system; a user may be referenced by zero or one schedule.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R4 monitoring_logs.type → settings.type

One parent may be referenced by zero to many child rows.
Each child must reference exactly one parent (NOT NULL foreign key).

R5 history_logs.type → settings.type

One parent may be referenced by zero to many child rows.
Each child must reference exactly one parent (NOT NULL foreign key).

R6 monitoring_logs.ph_reading_id → ph_readings.ph_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R7 monitoring_logs.ppm_reading_id → ppm_readings.ppm_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R8 notifications.ph_reading_id → ph_readings.ph_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R9 notifications.ppm_reading_id → ppm_readings.ppm_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

REFERENCE AND DELETION RULES

R1, R2, R3: deleting a user sets updated_by to NULL (ON DELETE SET NULL).

R4-R9: referenced parent rows cannot be deleted (ON DELETE RESTRICT).

settings.type is UNIQUE and can be referenced by a foreign key even though it is not the primary key.

monitoring_logs: ph_reading_id is only valid for type ph; ppm_reading_id is only valid for type ppm.

Monitoring and notification rows cannot reference both pH and PPM readings at once.

history_logs has no direct foreign key to readings or the Dithane schedule.

updated_by identifies only the last editor; it is not a complete audit trail.

ERD edges describe stored relationships; they do not establish a physical ESP32 connection.

Based on the project schema checked on 30 September 2026.